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One Electron at a Time: Coulomb Blockade, Single-Electron Transistors and Josephson Junctions

KKKC6353 Nanoelectronic Devices — Lecture 3, Document 3 of 3

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How to use this document

In Document 2, electrons tunnelled in huge numbers. This document shrinks the picture until single electrons count. When a piece of metal or semiconductor, an **island**, is only a few nanometres across, adding even one electron to it costs a noticeable amount of energy. That simple fact leads to:

- **Coulomb blockade**: electrons cannot tunnel onto a small island unless they are given enough energy to pay that cost;
- the **single-electron transistor** (SET), which lets electrons through one at a time and can sense a fraction of one electron's charge;
- **artificial atoms**, small dots whose electrons fill shells like the electrons of real atoms;
- and, when the metals are superconductors, the **Josephson junction**, the heart of the circuits that won the 2025 Nobel Prize in Physics and of many of today's quantum computers.

The document ends with a short comparison of the ordinary MOSFET, the SET and the carbon nanotube transistor, as a preview of Lecture 4.

As before, each section opens with a blue **Key ideas** box and closes with **check questions**, and the green **Going further** boxes hold the optional mathematics.

1 Charge comes in lumps

> **Key ideas**
>
> Electric charge always comes in whole electrons. You cannot add half an electron to anything.
>
> Adding one electron to a large object changes its energy by a negligible amount. Adding one to a nanometre-sized island changes it a lot.
>
> When that energy cost is larger than the thermal energy around it, single electrons start to matter.

1.1 A crowded lift

Think of a lift. If it is a huge goods lift, one more person stepping in makes no difference. If it is a tiny lift that is already full, one more person is a big deal: someone will have to squeeze, and it takes effort. Electrons on a piece of metal are similar. They all repel each other, so adding another one costs energy, because it must be pushed in against the repulsion of those already there.

How much energy depends on the size of the piece. For anything we can see, the cost of one extra electron is far too small to notice; charge seems to flow smoothly, like water. But for an island only a few nanometres across, the cost of one electron can be tens or hundreds of millielectronvolts. That is larger than the energy of the random thermal jiggling of electrons at room temperature, about 26 meV. And then charge no longer flows smoothly: it arrives in lumps, one electron at a time.

The cost of one extra electron is called the **charging energy**. It is set by the island's **capacitance**, C, a measure of how much charge it can hold for a given voltage. Smaller islands have smaller capacitance, and a higher charging energy.

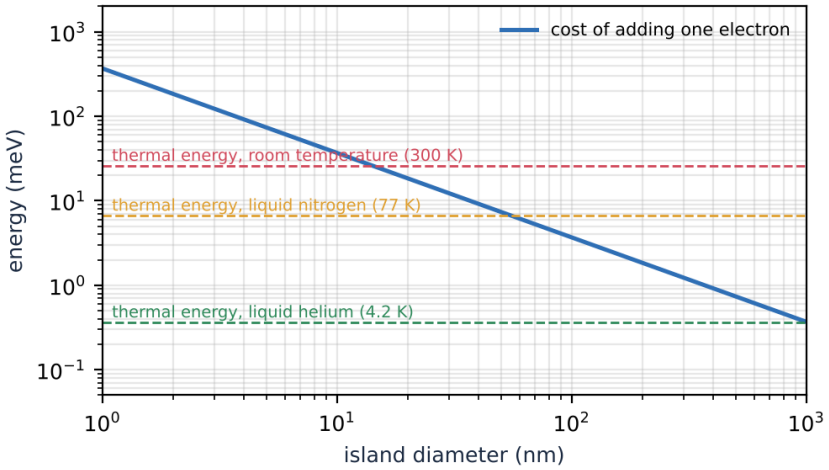


Figure 3.1. *The charging energy of a small island (here a sphere in silicon dioxide) against its diameter, compared with the thermal energy at three temperatures. Single-electron effects need the charging energy to be well above the thermal energy: an island of 100 nm works only at a few kelvin; one of a few nanometres can work at room temperature. Figure made for this course.*

> **Going further: the charging energy (optional)**
>
> Adding a charge e to an island of total capacitance C stores an energy
>
> $$E_C = \frac{e^2}{2C}.$$
>
> For a sphere of diameter d in a material of relative permittivity ε_r, far from other conductors, $C = 2\pi\varepsilon_0\varepsilon_r d$. In SiO_2 ($\varepsilon_r = 3.9$): $d = 100$ nm gives $E_C \approx 3.7$ meV; $d = 10$ nm gives 37 meV; $d = 2$ nm gives 185 meV.
>
> Compare these with the thermal energy $k_B T$: 26 meV at 300 K, 6.6 meV at 77 K, 0.36 meV at 4.2 K. $E_C = k_B T$ at room temperature for $d \approx 15$ nm. For clean single-electron behaviour we want E_C to be about ten times $k_B T$, which at room temperature needs islands of only 1 to 2 nm.
>
> *A note on conventions.* Some books call e^2/C the charging energy. That is the energy difference between adding the Nth and the $(N+1)$th electron, the **addition energy**. Both are correct as long as you say which you mean.

Check your understanding

1. Why does adding one electron to a small island cost more energy than adding one to a large one?
2. Why do single-electron effects in a 100 nm island disappear at room temperature?

2 Coulomb blockade

Key ideas

Connect a small island to a source and a drain through two tunnel junctions.

If the next electron would cost more energy than the voltage supplies, no electron can tunnel onto the island, and no current flows. This is Coulomb blockade.

It only works if the island is cold or small enough, and if the junctions are opaque enough that each electron really sits on the island.

2.1 Blocked

Put a small island between two metal leads, a **source** and a **drain**, separated from each by a thin insulating layer: a **tunnel junction**, exactly the kind of wall described in Document 1. Electrons can reach the drain only by tunnelling from the source onto the island, and then from the island to the drain.

Apply a small voltage between source and drain. An electron in the source is ready to tunnel onto the island, but if it did, the island's energy would rise by the charging energy. If the voltage does not provide that much energy, the electron cannot go. Neither can any other. Current is blocked, not by a lack of electrons nor by a thick wall, but by the repulsion of a single extra charge. This is **Coulomb blockade**, named after Coulomb's law of electric repulsion.

Raise the voltage above a threshold and the blockade lifts: one electron tunnels on, then off to the drain, then the next, in single file.

2.2 Two conditions

Coulomb blockade needs two things.

1. **The charging energy must beat the thermal energy.** Otherwise, a random thermal kick gives an electron the energy it needs anyway. So the island must be either very small (a few nanometres for room temperature) or very cold (below about 1 K for an island of a micrometre).
2. **The junctions must be opaque enough.** If electrons tunnel on and off too easily, no electron stays on the island long enough to be "an extra electron" there; its position is blurred by quantum uncertainty. This sets a minimum resistance for each tunnel junction, about 26 kΩ.

Going further: why 26 kilohms (optional)

An electron stays on the island for a time of about $\tau = RC$, where R is the junction's tunnelling resistance. By the energy–time uncertainty relation, its energy is then uncertain by about $\Delta E \approx h/\tau = h/RC$. For the charge to be well defined, this must be smaller than the addition energy e^2/C:

$$\frac{h}{RC} < \frac{e^2}{C} \quad \Rightarrow \quad R > \frac{h}{e^2} \approx 25.8\ \text{k}\Omega.$$

The quantity h/e^2 is the **resistance quantum**. It appears again in Section 6, in the quantum Hall effect.

Check your understanding

1. In your own words, what blocks the current in Coulomb blockade?
2. State the two conditions needed to see it, and explain each in one sentence.

3 The single-electron transistor

> **Key ideas**
>
> Add a gate next to the island, and you have a single-electron transistor (SET).
>
> The gate shifts the energy of the island's next available "place" up and down. When it lines up between the source and the drain, electrons flow one by one.
>
> As the gate voltage rises, the current turns on and off again and again: one peak for each extra electron on the island.

3.1 The structure

A **single-electron transistor** is the blocked island of Section 2 with one addition: a **gate** electrode close to the island but separated from it by an insulator too thick to tunnel through. It looks like a MOSFET in which the channel has been replaced by a tiny island with a tunnel junction at each end.

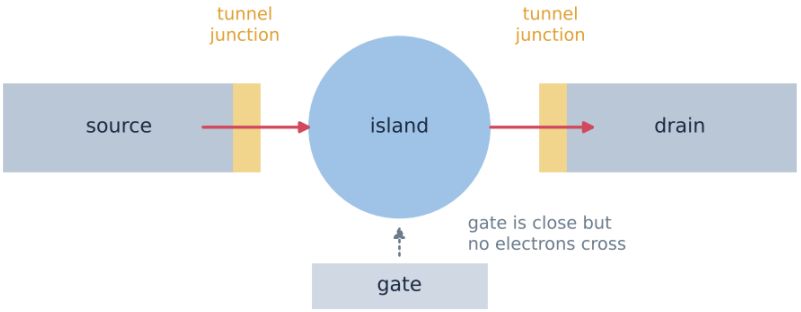


Figure 3.2. *A single-electron transistor. Electrons tunnel from the source onto the island and from the island to the drain, one at a time, through two tunnel junctions. The gate is close to the island but no electrons cross to it; it only changes the island's energy. Figure made for this course.*

3.2 How the gate controls it

Picture the island's electrons as filling a ladder of "places", each place one charging energy above the last. Normally the next empty place sits above the energy of the electrons in the source, and no electron can step onto it: the transistor is blocked.

The gate changes this. A positive voltage on the gate attracts electrons, which lowers the energy of the island's places. When the next empty place drops into the narrow window between the energies of the source and the drain, an electron tunnels on from the source, the place is filled, and then the electron tunnels off to the drain, emptying it again. The cycle repeats, and current flows, **one electron at a time**.

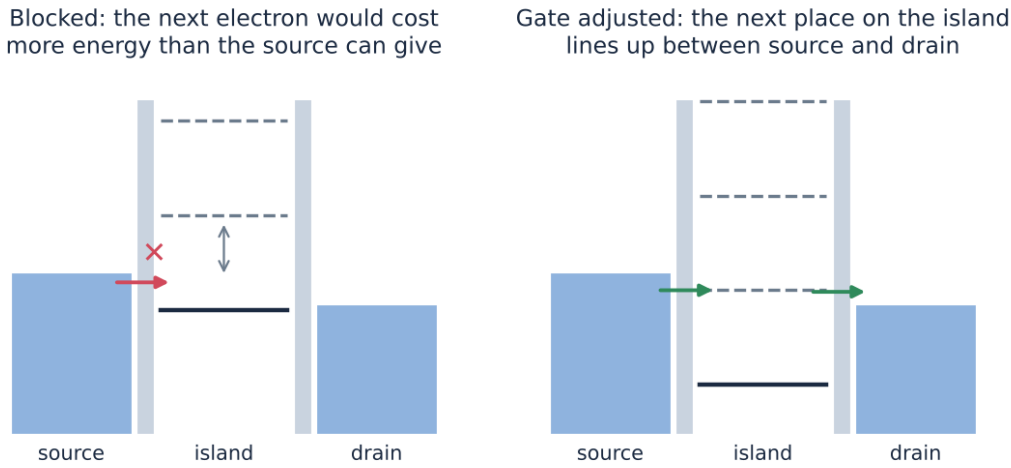


Figure 3.3. *Energy pictures of an SET. Shading shows the filled states of the source and drain; solid lines on the island are filled places, dashed lines empty ones. Left: the next empty place is above the source, so no electron can tunnel on; the current is blocked. Right: the gate has lowered the island's places until the next one lies between source and drain, and electrons pass through one by one. Figure made for this course; schematic.*

Raise the gate voltage a little more and that place drops below the drain. Now it stays filled; the island holds one more electron than before, and the current is blocked again. Keep raising the gate and the *next* place comes into the window, and so on.

3.3 Coulomb oscillations

So as the gate voltage rises steadily, the SET's current switches on and off again and again: sharp peaks, separated by flat regions where it is blocked. These are **Coulomb oscillations**. Between each pair of peaks the island holds a fixed whole number of electrons, N; each peak marks the change from N to $N + 1$. The peaks are evenly spaced, because each extra electron needs the gate to supply the same extra charge.

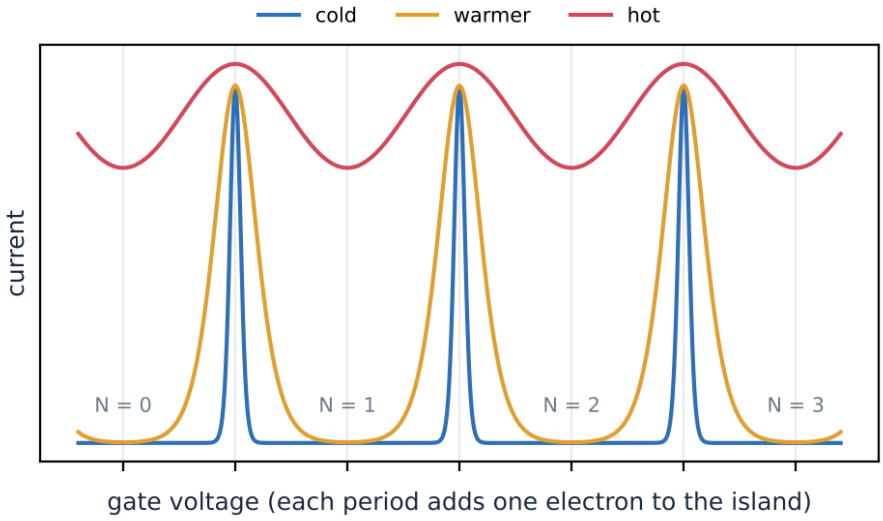


Figure 3.4. *Calculated current through an SET as the gate voltage rises, at three temperatures. Each peak lets the island gain one more electron (N = 0, 1, 2, 3 in the flat regions between). When the thermal energy is 1% of the charging energy the peaks are sharp; at 10% they broaden; at 40% they have almost washed out. Figure made for this course from the standard "orthodox" theory of single-electron tunnelling.*

Temperature is the enemy. As it rises, thermal kicks let electrons hop on and off even when they should be blocked, and the peaks blur into a gentle ripple.

3.4 Coulomb diamonds

An SET has two knobs: the gate voltage and the source–drain voltage. Plotting the current against both gives a striking picture. The blocked regions form a row of **Coulomb diamonds**. Inside each diamond the island holds a fixed number of electrons and no current flows; outside, current flows. The height of each diamond shows how much source–drain voltage is needed to force the next electron on, which directly measures the island's charging energy.

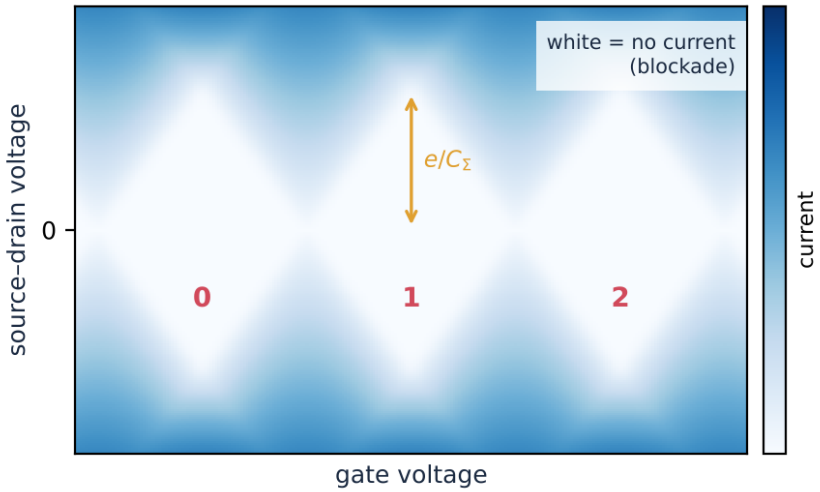


Figure 3.5. *Calculated current through an SET against gate voltage (across) and source–drain voltage (up). In the white diamonds the current is blocked and the island holds a fixed number of electrons (0, 1, 2). The height of a diamond is e/C_Σ, set by the island's total capacitance. Figure made for this course from the orthodox theory.*

> **Going further: the numbers in an SET (optional)**
> Let the island have total capacitance $C_\Sigma = C_1 + C_2 + C_g$, the sum of the capacitances to source, drain and gate. Then:

- The **gate period**, the change in gate voltage between neighbouring current peaks, is $\Delta V_g = e/C_g$. It adds exactly one electron's worth of charge to the gate capacitor.
- The **diamond height** (the largest source–drain voltage at which the current is still blocked) is e/C_Σ.
- The thermal width of each peak is a few $k_B T/e$ in source–drain voltage terms, which is why the peaks blur when $k_B T$ approaches E_C.

> The calculated figures in this section use the "orthodox theory": electrons tunnel one at a time, each tunnelling event is quick, and the rate of each event is set by the energy it releases. It describes metallic SETs very well.

Check your understanding

1. What does the gate of an SET do, if no electrons cross to it?
2. Why does the current through an SET rise and fall repeatedly as the gate voltage increases?
3. What happens to the Coulomb oscillations as the temperature rises, and why?

4. What does the island hold inside a Coulomb diamond?

Watch. NTT Basic Research Laboratories, *Silicon single-electron devices*, youtube.com/watch?v=-bsHP4DB3XQ.

Explore. The animation *Single-electron transistor* (faizverse.github.io/nanoelectronic-devices/L3) lets you turn the gate, watch electrons hop one at a time, warm the device up, and, in its "more detail" mode, draw the Coulomb diamonds.

4 Artificial atoms

> **Key ideas**
>
> In a semiconductor quantum dot, the island is small enough that its electrons also have quantised energy levels, as in Lecture 2.
>
> Adding an electron then costs the charging energy plus, sometimes, a jump to the next level.
>
> Dots with 2, 6 and 12 electrons are especially stable, like the noble gases among real atoms. These dots are "artificial atoms".

In a metal island, the electrons' own energy levels are so closely spaced that only the charging energy matters. In a semiconductor **quantum dot** of the kind described in Lecture 2, the levels are widely spaced, because the dot is small and the electrons are light. Then the cost of adding an electron has two parts: the charging energy, always; and, whenever the current level is full, the extra energy needed to start the next level.

Measuring the spacing of the Coulomb peaks therefore measures these levels. In 1996, Seigo Tarucha and colleagues in Japan and the Netherlands measured a small, flat, disc-shaped dot holding from 1 to about 20 electrons. The spacing was especially large after the 2nd, 6th and 12th electron. These are exactly the numbers at which electron shells close in a flat, two-dimensional "atom", just as helium, neon and argon are especially stable among real atoms. Smaller bumps after the 4th and 9th electron followed **Hund's rule**, the same rule that decides how electrons fill shells in a real atom.

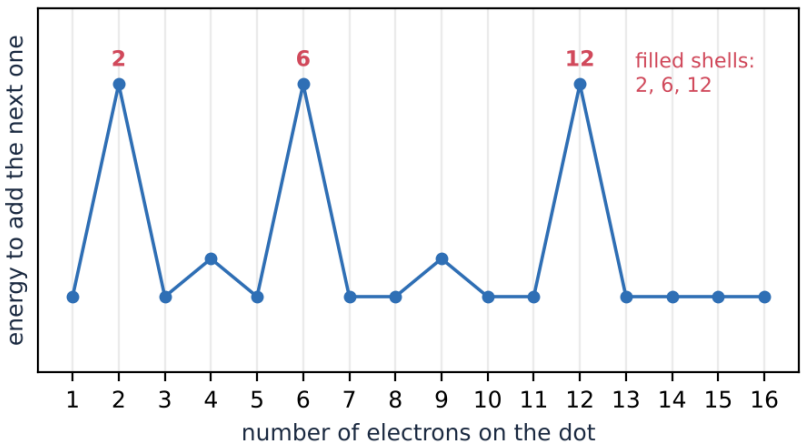


Figure 3.6. *The energy needed to add the next electron to a small disc-shaped quantum dot, against the number already there. Extra energy is needed after 2, 6 and 12 electrons, when a shell is full; smaller bumps at 4 and 9 come from Hund's rule. Figure made for this course; schematic, after the measurements of Tarucha et al. (1996).*

So a quantum dot is an **artificial atom**: a few electrons, held not by a nucleus but by a man-made trap, filling shells by the rules of quantum mechanics. Unlike a real atom, its size and shape, and so its "chemistry", can be set by design and tuned with a gate.

Check your understanding

1. Why do the Coulomb peaks of a semiconductor quantum dot not have equal spacing, unlike those of a metal island?
2. What is special about a quantum dot holding 2, 6 or 12 electrons?

5 What single-electron devices are good for

> **Key ideas**
> SETs are not going to replace the transistors in computer chips.
> They are, however, the most sensitive detectors of electric charge ever made, and they read out the qubits in silicon quantum computers.
> Devices that move exactly one electron per cycle now help define the ampere.

5.1 Not a replacement for the MOSFET

An SET switches with a single electron, so it seemed in the 1990s that it might become the ultimate low-power transistor. Several problems have stood in the way:

- **Room temperature needs islands of 1 to 2 nm**, and every island in a chip would have to be made to that size with atomic precision. Islands this small have been made and shown to work at room temperature in the laboratory, but not by the billion.
- **The current is tiny**, a few nanoamperes at most, because each electron must tunnel through two high-resistance junctions. Circuits built from SETs are slow and cannot drive long wires.
- **SETs are too sensitive.** A single stray charge trapped in the insulator nearby shifts the Coulomb oscillations. Such charges hop around randomly, and every SET on a chip would be shifted differently.

5.2 The most sensitive charge detector

The last problem is also an SET's greatest strength. Bias the SET halfway up the side of a Coulomb peak, where the current changes most steeply with gate voltage. Now the tiniest change in the charge near the island, a small fraction of one electron, shifts the peak and produces a large change in current. An SET is an **electrometer** of unrivalled sensitivity.

This is how many **quantum computers** read their qubits. In silicon spin-qubit processors, each qubit is a single electron in a quantum dot, with its quantum information stored in the direction of the electron's spin. Measuring a spin directly is very hard, so the spin is first converted into a charge (whether the electron can move to a neighbouring dot or not), and a nearby single-electron sensor detects that charge.

5.3 Counting electrons: a new ampere

A related device, the **single-electron pump**, uses gates to move exactly one electron through a chain of islands for each cycle of an applied signal. At a frequency f it delivers a current $I = ef$: count the cycles, and you have counted the electrons. Silicon and gallium arsenide pumps run at around a gigahertz, giving currents of about 0.16 nanoamperes, accurate to about one part in a million or better.

In 2019 the world's system of units, the SI, was redefined so that the charge of the electron is now an exact, fixed number: $e = 1.602\,176\,634 \times 10^{-19}$ coulombs. The ampere is defined from it. Electron pumps are one way of putting that definition into practice. With two other quantum effects, they form the **quantum metrology triangle**, which links current, voltage and resistance through fundamental constants alone.

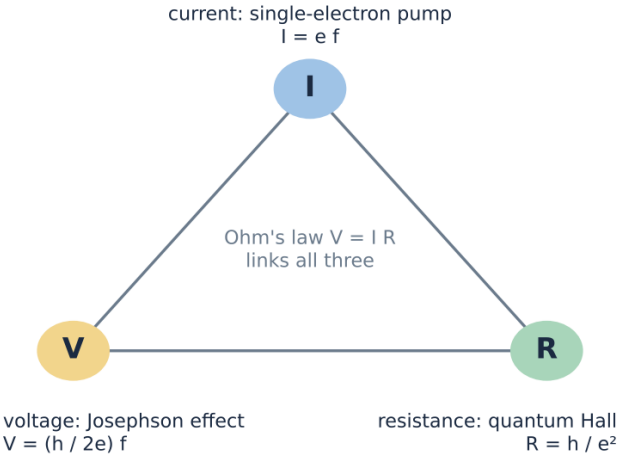


Figure 3.7. *The quantum metrology triangle. Single-electron pumps give a current from a frequency and the electron charge; the Josephson effect (Section 6) gives a voltage from a frequency; the quantum Hall effect gives a resistance from h/e^2. Ohm's law links all three, so measuring around the triangle tests our understanding of all three effects. Figure made for this course.*

Check your understanding

1. Give two reasons why SETs have not replaced MOSFETs in computer chips.
2. Why is an SET such a sensitive detector of charge? Name one place where that is used today.
3. How does a single-electron pump produce a known current?

6 Tunnelling pairs: the Josephson junction

> **Key ideas**
>
> In a superconductor, electrons pair up and the pairs move together without resistance.
>
> The pairs can tunnel through a thin insulator between two superconductors, carrying current with no voltage at all. This is the Josephson effect.
>
> In 1984–85, experiments showed a whole circuit made of billions of electrons tunnelling as a single quantum object. This won the 2025 Nobel Prize in Physics and underlies today's superconducting quantum computers.

6.1 Superconductors and pairs

Some metals, such as aluminium and niobium, lose all electrical resistance when cooled below a critical temperature, about 1.2 K for aluminium and 9 K for niobium. In this **superconducting** state the electrons form **Cooper pairs**, and all the pairs act together as a single quantum wave stretching across the whole piece of metal.

6.2 The Josephson effect

In 1962 Brian Josephson, then a 22-year-old student at Cambridge, predicted that Cooper pairs should tunnel through a thin insulating layer, 1 to 2 nm thick, between two superconductors, and carry a current with **no voltage** across the junction at all. If a voltage V is applied, the current instead oscillates at a frequency $f = 2eV/h$, about 484 GHz per millivolt. Both predictions were soon confirmed, and Josephson shared the 1973 Nobel Prize with Esaki and Giaever.

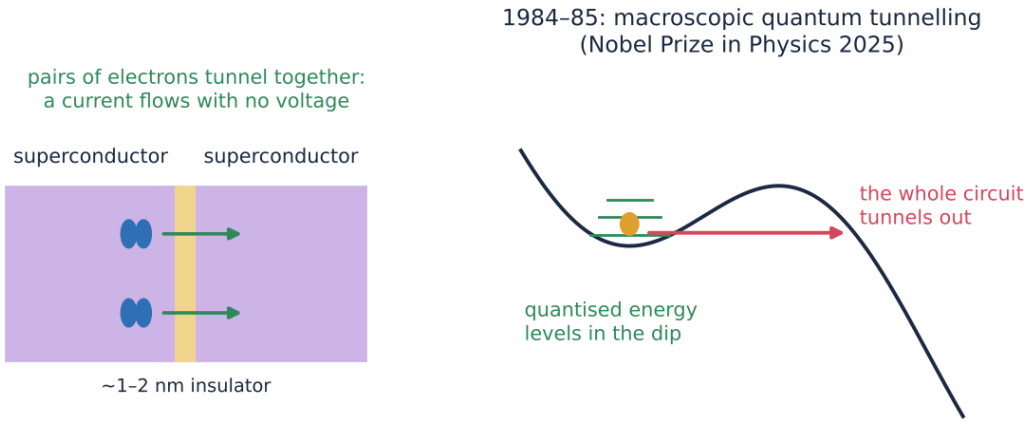


Figure 3.8. *Left: a Josephson junction, two superconductors separated by a thin insulator. Cooper pairs tunnel through together, and a current flows with no voltage. Right: the circuit's state sits in a dip of a tilted "washboard" of energy, with quantised energy levels. In the 1984–85 experiments, the whole circuit tunnelled out of the dip through the barrier. Figure made for this course; schematic.*

The Josephson effect is the most precise link between voltage and frequency we have. Arrays of thousands of junctions, driven at a known microwave frequency, produce exact voltages and are the world's voltage standards: the "V" corner of the metrology triangle. Josephson junctions in loops, called **SQUIDs**, are the most sensitive detectors of magnetic fields, used to record the faint fields of the human brain.

6.3 Macroscopic quantum tunnelling: the 2025 Nobel Prize

Everything in this lecture so far has involved the tunnelling of individual particles. Could something large, a whole electrical circuit containing billions of electrons, tunnel as a single object? In 1984 and 1985, John Clarke, Michel Devoret and John Martinis, at the University of California, Berkeley, showed that it can.

They passed a current through a Josephson junction just below the value at which a voltage appears. The state of the whole circuit then sits in a shallow dip of energy, like a ball in a dip of a tilted washboard. Classically it should stay there until a thermal kick lifts it out. But at the lowest temperatures, it still escaped, at a rate that stopped depending on temperature: the circuit was **tunnelling** out through the energy barrier. By shining microwaves on it they also showed that the circuit had **quantised energy levels** in the dip, exactly as a single atom or a particle in a box does.

In other words, an electrical circuit large enough to see with the naked eye behaved as one quantum particle. They received the **2025 Nobel Prize in Physics** "for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit".

The consequences are all around us in quantum technology. A circuit with quantised energy levels can be used as a **qubit**: its lowest two levels stand for 0 and 1. The superconducting quantum processors built by Google, IBM and others are made of exactly such Josephson-junction circuits. Martinis himself led the Google team whose 53-qubit processor performed its first "beyond classical" computation in 2019.

Check your understanding

1. What tunnels through a Josephson junction, and what is unusual about the current it carries?
2. What did Clarke, Devoret and Martinis show that was new? Why is it called *macroscopic* quantum tunnelling?
3. How is that discovery connected to quantum computers?

7 How single-electron devices are made

> **Key ideas**
>
> The classic SET is made of aluminium, by evaporating metal at two angles through a suspended mask so that two layers just overlap.
>
> The overlap, separated by a 1–2 nm layer of aluminium oxide, forms a tunnel junction.
>
> Silicon SETs and quantum dots are made with transistor-style processing; tiny islands can also be chemically made nanoparticles.

Shadow evaporation. The first SET, made by Theodore Fulton and Gerald Dolan at Bell Labs in 1987, used a trick still used for most superconducting qubits today. Electron-beam lithography patterns a resist with a narrow suspended "bridge". Aluminium is evaporated from one angle, then the surface is exposed to oxygen to grow a 1 to 2 nm layer of aluminium oxide. Then aluminium is evaporated again from a different angle. The bridge casts a different shadow each time, so the two aluminium layers are shifted and just overlap. Where they overlap, separated by the oxide, is a tunnel junction. One evaporation

sequence gives junctions with areas of about 100 nm by 100 nm, which work at temperatures below about 1 K.

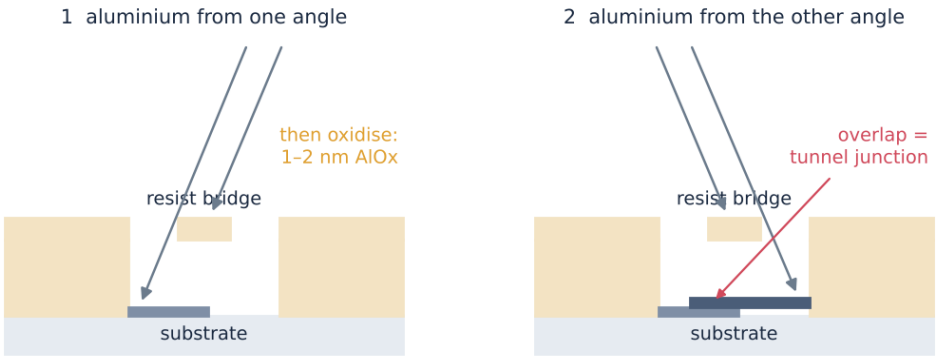


Figure 3.9. *Shadow evaporation (the Dolan bridge technique). Aluminium is evaporated through a suspended resist bridge from two angles, with an oxidation step between. The two layers overlap under the bridge, forming a tunnel junction. Figure made for this course; schematic.*

Silicon. Silicon SETs and quantum dots are made with the processes of Lecture 1: thin silicon wires or fins, with gates that create barriers electrically, so that the island and its tunnel junctions are defined by voltages rather than by layers of insulator. The silicon spin-qubit chips of Section 5 are made this way, some on industrial 300 mm production lines.

Nanoparticles and molecules. For islands only a few nanometres across, small enough for room temperature, chemists can make metal or semiconductor nanoparticles (Lecture 2, Document 3) and trap one in a narrow gap between two electrodes. The difficulty is doing so reliably.

Check your understanding

1. In shadow evaporation, what forms the tunnel barrier, and how is it made?
2. Why do room-temperature SETs require such different methods from the aluminium SETs used at low temperature?

8 Preview: three transistors compared

> **Key ideas**
>
> The MOSFET, the SET and the carbon nanotube FET all have a source, a drain and a gate, but they work in very different ways.
>
> This table is a preview. Lecture 4 treats the carbon nanotube FET in detail.

The table below sets the transistor you know from Lecture 1 beside the SET of this document and the carbon nanotube transistor (CNTFET) introduced in Lecture 2. It is deliberately simple. Before Lecture 4, try sketching a band diagram for each, from source to drain, with the gate off and on.

	MOSFET	Single-electron transistor	Carbon nanotube FET
Channel	A thin layer of silicon	A tiny island between two tunnel junctions	One or more semiconducting carbon nanotubes

	MOSFET	Single-electron transistor	Carbon nanotube FET
How electrons travel	Over an energy barrier set by the gate	Tunnelling on and off the island, one at a time	Through the tube, often without scattering; across the contacts by tunnelling or over a barrier
What the gate does	Raises or lowers the barrier	Shifts the island's energy, one electron's worth per period	Raises or lowers the barrier in the tube, and thins the contact barriers
Current against gate voltage	Turns on once, then stays on	Peaks repeatedly (Coulomb oscillations)	Turns on once, like a MOSFET
Room temperature?	Yes	Only with islands of 1–2 nm	Yes
Current	Large	Very small (nanoamperes)	Large per tube, but tubes are thin
Status in 2026	Every computer chip	Charge sensors, qubit readout, metrology	Research prototypes, including a small 16-bit processor (2019)

Check your understanding

1. Which of the three transistors gives a current that rises and falls repeatedly as the gate voltage increases, and why?
2. In which of the three does tunnelling play a central part?

9 Summary, and what comes next

1. **Charge comes in whole electrons.** Adding one to a tiny island costs the charging energy $e^2/2C$, which grows as the island shrinks.
2. **Coulomb blockade**: when that cost exceeds what the voltage or the heat can supply, no electron can tunnel onto the island. It needs small or cold islands and junctions of more than about 26 kΩ.
3. **The single-electron transistor** uses a gate to shift the island's energy; its current peaks once for every electron added, giving Coulomb oscillations and Coulomb diamonds.
4. **Small semiconductor dots are artificial atoms**, filling shells of 2, 6 and 12 electrons.
5. **SETs will not replace MOSFETs**, but they are the most sensitive charge detectors, read out spin qubits, and as electron pumps help realise the ampere.
6. **Cooper pairs tunnel through Josephson junctions**, giving exact voltage standards and SQUIDs.
7. **A whole circuit can tunnel as one quantum object**, as Clarke, Devoret and Martinis showed in 1984–85, which won the 2025 Nobel Prize in Physics and underlies superconducting quantum computers.

Where Lecture 4 begins. Lecture 4 looks closely at the transistors that may follow, or extend, today's silicon: carbon nanotube transistors, FinFETs and silicon-on-insulator, and compound semiconductors.

Words to know

Word	Plain meaning
Island	A tiny conductor connected to the outside only through tunnel junctions
Charging energy	The energy cost of adding one electron to an island, $e^2/2C$
Coulomb blockade	No current flows because adding one electron costs too much energy
Single-electron transistor (SET)	An island between two tunnel junctions, with a gate to control its energy
Coulomb oscillations	Repeated current peaks as the gate voltage of an SET rises
Coulomb diamonds	Diamond-shaped blocked regions in a plot of SET current against two voltages
Artificial atom	A quantum dot whose electrons fill shells like those of an atom
Electron pump	A device that moves exactly one electron per cycle
Cooper pair	Two electrons bound together in a superconductor
Josephson junction	Two superconductors separated by a thin insulator, through which pairs tunnel
Macroscopic quantum tunnelling	A large object, such as a whole circuit, tunnelling as one quantum particle
Qubit	The basic unit of a quantum computer, a system with two quantum states

Sources and further reading

Gentle introductions

- NTT Basic Research Laboratories, *Silicon single-electron devices*, youtube.com/watch?v=-bsHP4DB3XQ.
- The Royal Swedish Academy of Sciences, *Popular science background: Nobel Prize in Physics 2025*, nobelprize.org.

Textbooks and reviews

- K. K. Likharev, "Single-electron devices and their applications", *Proceedings of the IEEE* 87, 606 (1999).
- L. P. Kouwenhoven, D. G. Austing and S. Tarucha, "Few-electron quantum dots", *Reports on Progress in Physics* 64, 701 (2001).
- J. P. Pekola et al., "Single-electron current sources: toward a refined definition of the ampere", *Reviews of Modern Physics* 85, 1421 (2013).

Original work

- T. A. Fulton and G. J. Dolan, "Observation of single-electron charging effects in small tunnel junctions", *Physical Review Letters* 59, 109 (1987).
- S. Tarucha, D. G. Austing, T. Honda, R. J. van der Hage and L. P. Kouwenhoven, "Shell filling and spin effects in a few electron quantum dot", *Physical Review Letters* 77, 3613 (1996).

- B. D. Josephson, “Possible new effects in superconductive tunnelling”, *Physics Letters* 1, 251 (1962).
- M. H. Devoret, J. M. Martinis and J. Clarke, “Measurements of macroscopic quantum tunneling out of the zero-voltage state of a current-biased Josephson junction”, *Physical Review Letters* 55, 1908 (1985).
- J. M. Martinis, M. H. Devoret and J. Clarke, “Energy-level quantization in the zero-voltage state of a current-biased Josephson junction”, *Physical Review Letters* 55, 1543 (1985).

Credit. Sections 2 to 4 draw on F. J. Himpsel’s lecture slides on single-electron transistors (University of Wisconsin–Madison), used in earlier years of this course and now archived.

Figures. All figures were made for KKKC6353 (scripts in `figures-source/`). The Coulomb oscillations and diamonds were calculated with the orthodox theory of single-electron tunnelling.